

2.3 Exploration Tasks

I. Document the schema of each file

Boston, Massachusetts, United States - 27 December, 2025

- listings.csv

Column name	Data types	Ranges	Sample values
id	integer	0 - ∞	3781, 5506
name	string		Harborside-Walk to subway, Fort Hill Inn Private! Minutes to center!
host_id	integer	0 - ∞	4804, 8229
host_profile_id	integer	0 - ∞	1.46E+18, 1.46E+18
host_name	string		Frank, Terry
neighbourhood_group	-	-	NULL
neighbourhood	string	-	East Boston, Roxbury
latitude	float	-90 to 90	42.36413
longitude	float	-180 to 180	-71.0299
room_type	string	entire home/apt / private room	Entire home/apt
minimum_nights	integer	0 - ∞	29, 3
number_of_reviews	integer	0 - ∞	27, 140
last_review	date		9/30/2025
reviews_per_month	float	0 - ∞	0.21
calculated_host_listings_count	integer	0 - ∞	1, 11
availability_365	integer	0 - 365	184, 298
number_of_reviews_ltm	integer	0 - ∞	1, 9
license	integer		STR-490093

- listings.csv.gz

Column name	Data types	Ranges	Sample values
id	integer	0 - ∞	3781, 5506
listing_url	string		https://www.airbnb.com/rooms/3781
scrape_id	big integer	0 - ∞	2.03E+13
last_scraped	datetime		12/27/2025, 12/27/2025
source	string	city scrape	City scrape
name	string		Harborside-Walk to subway, Fort Hill Inn Private! Minutes to center!
description	string		Comfortable, Fully Equipped private apartment on 2 levels. Sleeps up to 4. Sleeper sofa in living room, Spiral Steps inside unit to bedroom.
neighborhood_overview	-	-	NULL

picture_url	string		https://a0.muscache.com/pictures/hosting/Hosting-3781/original/4222dd36-0b3f-4fc5-afb0-264cd76932b9.jpeg
host_id	integer	0 - ∞	4804, 8229
host_url	string		https://www.airbnb.com/users/show/4804
host_profile_id	integer		1.46E+18, 1.46E+18
host_profile_url	string		https://www.airbnb.com/users/profile/1462506429675641583
host_name	string		Frank, Terry
host_since	date	-	NULL
host_time_as_user_years	integer		17, 16
hosts_time_as_user_months	integer		0, 10
hosts_time_as_host_years	integer		14, 11
hosts_time_as_host_months	integer		11, 10
host_location	string		Massachusetts, United States, Boston, MA
host_about	string		My wife and I and grown children frequently occupy the spaces that we rent and as such try to make the spaces
host_response_time	-		
host_response_rate	-		
host_acceptance_rate	-		
host_is_superhost	boolean	t or f	T, f
host_thumbnail_url	-	-	NULL
host_picture_url	string		https://a0.muscache.com/im/users/4804/profile_pic/1327953150/original.jpg?aki_policy=profile_x_medium
host_neighbourhood	string	-	NULL
host_listings_count	integer		3, 12
host_total_listings_count	-	-	NULL
host_verifications	string		
host_has_profile_pic	boolean	t or f	T, f
host_identity_verified	boolean	t or f	T, f
neighbourhood	-	-	NULL
neighbourhood_cleansed	string		East Boston, Roxbury
neighbourhood_group_cleansed	-		
latitude	float	-90 to 90	42.36413
longitude	float	-180 to 180	-71.0299
property_type	string	selected range of strings	Entire rental unit, Entire guest suite, Entire condo, Private room in home
room_type	string	entire home/apt / private room	Entire home/apt

accommodates	integer		2, 4
bathrooms	float		1, 1.5
bathrooms_text	string	selected range of strings	1 bath
bedrooms	integer		1, 5
beds	integer		1, 3,
amenities	list		["Dishes and silverware", "Coffee maker", ..]
price	-		NULL
minimum_nights	integer		29, 3
maximum_nights	integer	max 1125	1125, 730
minimum_minimum_nights	integer		29, 3
maximum_minimum_nights	integer		29,5
minimum_maximum_nights	integer		1125, 730
maximum_maximum_nights	integer		1125, 28
minimum_nights_avg_ntm	float		29, 10.1
maximum_nights_avg_ntm	float		1125, 120
calendar_updated	-		NULL
has_availability	binary	t or f	T, F
availability_30	integer	0-30	29, 5
availability_60	integer	0-60	59, 23
availability_90	integer	0-90	89,53
availability_365	integer	0-365	184, 298
calendar_last_scraped	date		12/27/2025
number_of_reviews	integer		27, 140
number_of_reviews_ltm	integer		1, 10
number_of_reviews_l30d	integer		0, 1, 2
availability_eoy	integer		4, 5
number_of_reviews_ly	integer		1, 10, 0
estimated_occupancy_l365d	integer		58, 54
estimated_revenue_l365d	-		NULL
first_review	date		7/10/2015
last_review	date		9/30/2025
review_scores_rating	float		4096
review_scores_accuracy	float		5
review_scores_cleanliness	float		4.96
review_scores_checkin	float		5
review_scores_communication	float		4.96
review_scores_location	float		4.85
review_scores_value	float		4.89
license	string		STR-490093
instant_bookable	-		NULL
calculated_host_listings_count	integer		1, 11
calculated_host_listings_count_entire_homes	integer		1,11
calculated_host_listings_count_private_rooms	integer		0, 39
calculated_host_listings_count_shared_rooms	integer		0
reviews_per_month	float		0.21, 0.69

- [calendar.csv.gz](#)

Column name	Data types	Ranges	Sample values
listing_id	integer	0 - ∞	3781, 5506
date	date		12/27/2025
available	boolean	t or f	T, f
price	-	-	NULL
adjusted_price	-	-	NULL
minimum_nights	integer		29, 3
maximum_nights	integer	0 - 1125	1125,

- [reviews.csv.gz](#)

Column name	Data types	Ranges	Sample values
listing_id	integer	0 - ∞	3781, 5506
id	integer	0 - ∞	37776825
Date	date		8/9/2015
reviewer_id	integer	0 - ∞	36059247
reviewer_name	string		Greg, Damien
comments	string		The apartment was as advertised and Frank was incredibly helpful through the entire process. I would definitely recommend this place.

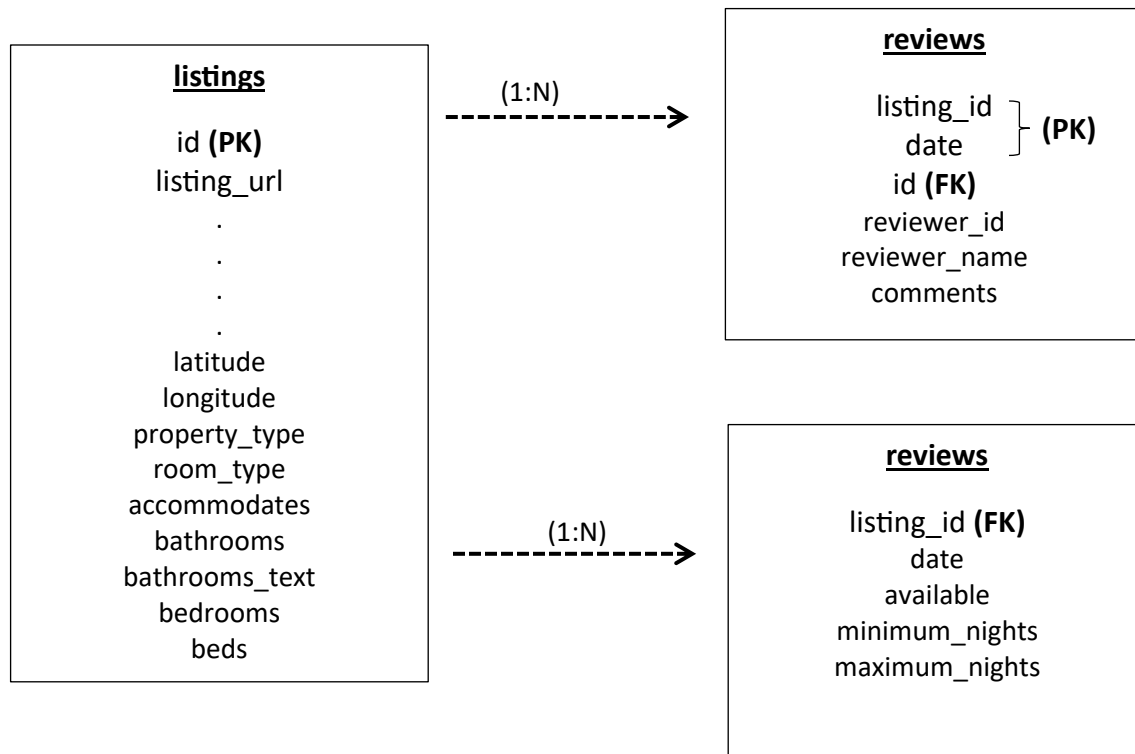
- [reviews.csv](#)

Column name	Data types	Ranges	Sample values
listing_id	integer	0 - ∞	3781, 5506
date	date		12/27/2025

- [neighbourhoods.csv](#)

Column name	Data types	Ranges	Sample values
neighbourhood_group	-		NULL
neighbourhood	string		Back Bay, Allston

II. Map the relationships between files and identify primary and foreign keys where applicable



III. Review the official data dictionary and note any columns requiring special interpretation.

- **price** - Stored as a text string with currency symbols (\$1,250.00). Must be stripped of non-numeric characters before performing any mathematical calculations.
- **available** - Stored as 't' or 'f'. For the calendar file, 'f' (false) is assumed to mean the day is occupied or booked, allowing us to compute occupancy rates and revenue estimates.

IV. Identify dataset limitations

- Because this data is extracted via automated web scrapers, unstructured free-text fields (such as description or amenities) often contain broken HTML formatting characters, trailing escape keys (`\n`, `\t`, `<b>`), or messy string variations ("Private Room", "private room", "PRIVATE ROOM "). This requires standardizing all text inputs to lowercase and trimming whitespaces before performing grouping operations.
- The dataset only includes active listings visible on the platform at the time of the scrape. If a highly popular property was permanently deactivated or deleted a month before the scrape occurred, its entire history is missing from the dataset.

- The calendar file does not show past bookings; it provides a 365-day forward-looking projection of availability. Using it to estimate historical revenue assumes that future booking patterns match past performance, which may not capture seasonal shifts.

V. Describe the business domain context: what does each entity (listing, host, review) represent

To extract meaningful insights from this data, we must clearly define what each real-world entity represents within the short-term rental ecosystem

- The Listing Entity: This represents a specific piece of physical real estate offered for rent on the platform. It is the central engine of revenue generation. It is defined by physical attributes (property type, room format, bed counts, square footage) and operational attributes (nightly price, minimum stay rules, geographic location).
- The Host Entity: This represents the individual or corporate property management firm operating the listing. Hosts control pricing strategies, set calendar availability, and handle guest interactions.
- The Review Entity: This represents user-generated feedback left by travelers after a completed stay. Because platforms only allow reviews following a verified booking, the frequency and volume of these entities serve as an excellent proxy for historical transaction volume, market popularity, and customer satisfaction.